

Suggested Answers for 2017 H2 Chemistry (9729) Prelim Paper 3

- 1 (a) (i) Group 2 elements have giant metallic structures with strong electrostatic attraction between a lattice of positive ions and delocalised electrons.

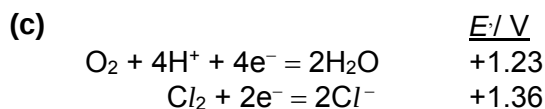
Since Ca^{2+} has a smaller radius and thus greater charge density than Ba^{2+} , more energy is required to overcome the stronger metallic bonds in Ca than that in Ba. Hence Ca has a higher melting point than Ba.

- (ii) Solution turns from purple to blue OR effervescence is observed with calcium and barium. White precipitate is observed with calcium only.



Ca has a larger nuclear charge and smaller radius than K while shielding effect by inner shell electrons is relatively similar. Thus the nuclear attraction on the outermost electron is stronger in Ca as compared to that in K. Hence the first ionisation energy of calcium is higher than that of potassium.

The 4s electron to be removed from Ca is nearer to the nucleus and has lower energy than the 4p electron to be removed from Ga. Hence the first ionisation energy of calcium is higher than that of gallium.

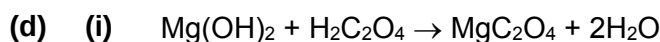


For *dilute* $\text{BaCl}_2(\text{aq})$,

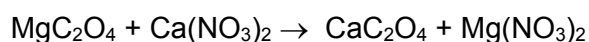
H_2O is more easily oxidised than Cl^- since $E(\text{O}_2/\text{H}_2\text{O})$ is less positive than $E(\text{Cl}_2/\text{Cl}^-)$. Hence, O_2 gas is produced at the anode.

For *concentrated* $\text{BaCl}_2(\text{aq})$,

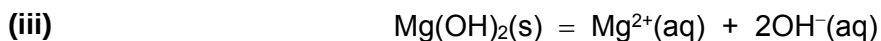
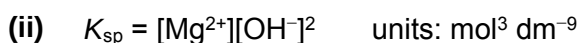
Higher $[\text{Cl}^-]$ causes the position of equilibrium of $\text{Cl}_2 + 2\text{e}^- = 2\text{Cl}^-$ to shift left, making $E(\text{Cl}_2/\text{Cl}^-)$ less positive than $E(\text{O}_2/\text{H}_2\text{O})$. Thus, Cl^- becomes more easily oxidised than H_2O and hence, Cl_2 gas is produced at the anode.



A



B



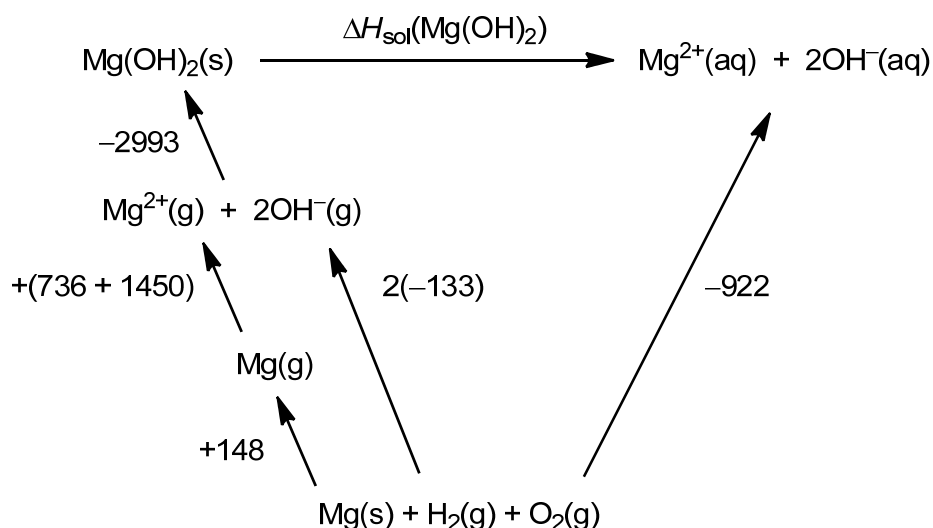
$$2.00 \times 10^{-11} = (x)(2x)^2 = 4x^3$$

$$[\text{Mg}(\text{OH})_2(\text{aq})] = x = 1.71 \times 10^{-4} \text{ mol dm}^{-3}$$

$$[\text{OH}^-] = 2x = 2 \times 1.71 \times 10^{-4} = 3.42 \times 10^{-4} \text{ mol dm}^{-3}$$

$$\text{pH} = 14 - (-\lg 3.42 \times 10^{-4}) = 10.5$$

(e)



By Hess' Law,

$$\begin{aligned}
 \Delta H'_{\text{sol}}(\text{Mg(OH)}_2(\text{s})) &= -(-2993) - (736 + 1450) - (148) - 2(-133) + (-922) \\
 &= +3 \text{ kJ mol}^{-1}
 \end{aligned}$$

- 2 (a) Reactivity decreases down the group.

$$E(\text{Cl}-\text{Cl}) = +244 \text{ kJ mol}^{-1}, E(\text{Br}-\text{Br}) = +193 \text{ kJ mol}^{-1}; E(\text{I}-\text{I}) = +151 \text{ kJ mol}^{-1}$$

$$E(\text{H}-\text{Cl}) = +431 \text{ kJ mol}^{-1}, E(\text{H}-\text{Br}) = +366 \text{ kJ mol}^{-1}; E(\text{H}-\text{I}) = +299 \text{ kJ mol}^{-1}$$

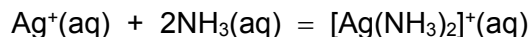
This is because the decrease in $E(\text{H}-\text{X})$ is more significant than the decrease in $E(\text{X}-\text{X})$ down the group.

- (b) (i) BaSO_4

- (ii) Br_2 causes a larger increase in the oxidation state of S from +2 in $\text{S}_2\text{O}_3^{2-}$ to +6 in SO_4^{2-} , as compared to I_2 which causes a smaller increase in the oxidation state of S from +2 in $\text{S}_2\text{O}_3^{2-}$ to +2.5 in $\text{S}_4\text{O}_6^{2-}$.

Therefore, Br_2 is a stronger oxidising agent than I_2 .

- (c) When $\text{NH}_3(\text{aq})$ is added, $[\text{Ag}(\text{NH}_3)_2]^+$ complex ion is formed.

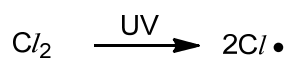


Due to the formation of this complex ion, $[\text{Ag}^+]$ in the solution decreases and causes the position of equilibrium of $\text{AgX}(\text{s}) = \text{Ag}^+(\text{aq}) + \text{X}^-(\text{aq})$ to shift to the right, increasing the solubility of both AgX .

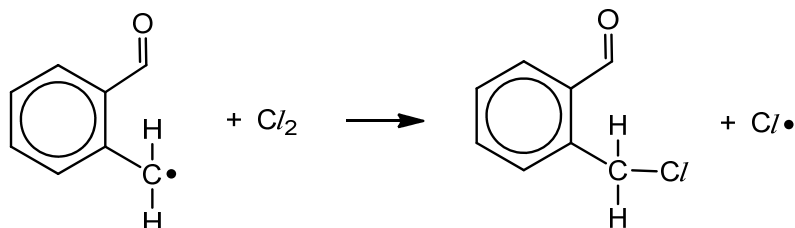
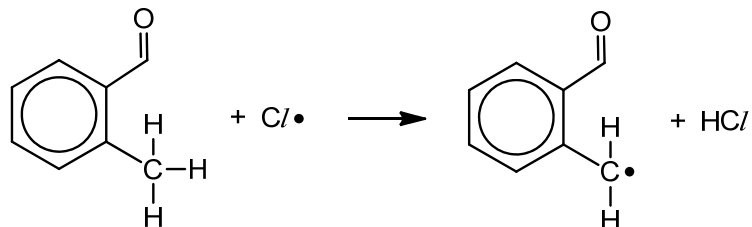
However, AgCl has a higher K_{sp} than AgI so AgCl dissolves completely but AgI hardly dissolves in $\text{NH}_3(\text{aq})$.

(d) (i) Name of mechanism: free radical substitution

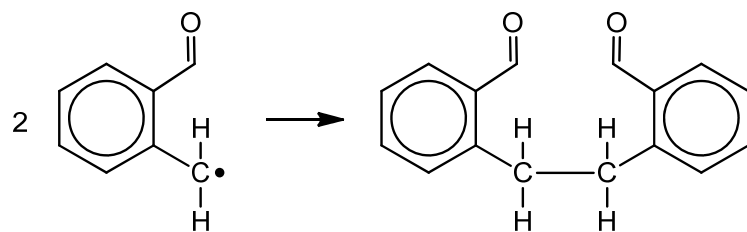
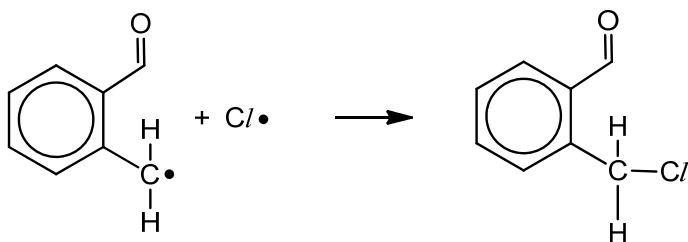
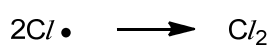
Initiation step



Propagation steps



Termination steps



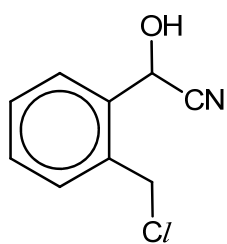
any 2 equations

(ii) II: HCN, trace amount of NaCN or NaOH

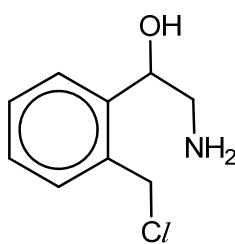
III: LiAlH₄, dry ether

IV: ethanol, heat in sealed tube

(iii)

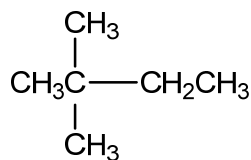
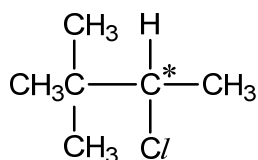


D



E

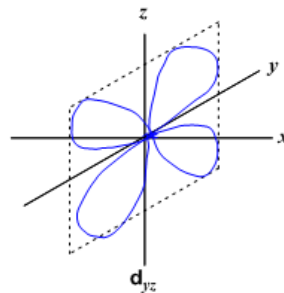
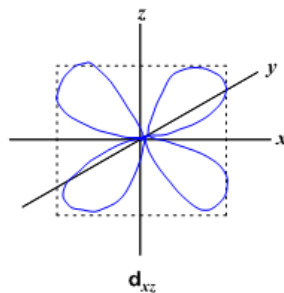
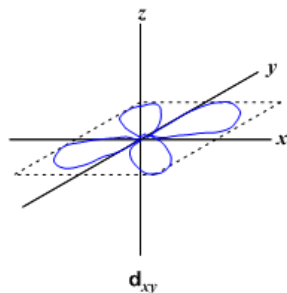
(iv)

**G****H**

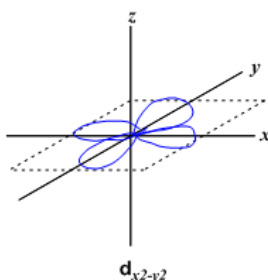
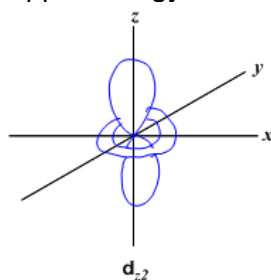
H has a chiral C and thus exhibits enantiomerism.

3 (a) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$

(b) (i) Lower energy level:



Upper energy level:



(ii) Chromium complexes are coloured due to the electron transition between d -orbitals.

The presence of ligands causes the d orbitals to split into 2 different energy levels.

The energy gap is relatively small such that radiation from the visible light is absorbed when an electron transits from a lower energy d -orbital to higher energy d -orbital which is partially filled.

Hence, the colour seen is the complement of the colours absorbed.

(c) (i) In alkaline medium, OH^- reacts with H^+ which reduces $[\text{H}^+]$. This causes the position of equilibrium of $\text{Cr}_2\text{O}_7^{2-} + \text{H}_2\text{O} \rightleftharpoons 2\text{CrO}_4^{2-} + 2\text{H}^+$ to shift right. Hence, there will be more CrO_4^{2-} in the mixture than $\text{Cr}_2\text{O}_7^{2-}$.

(ii) $\text{CrO}_4^{2-} + 4\text{H}_2\text{O} + 3\text{e}^- \rightarrow \text{Cr}(\text{OH})_3 + 5\text{OH}^-$

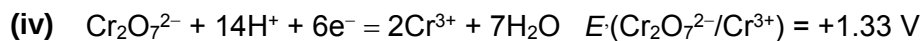
(iii)	mole ratio of	Cr(OH)_3	:	e^-	:	reducing agent
		0.200	:	$3(0.200)$		
				$3(0.200)$	:	0.300
		2	:	6	:	3

Using $\Delta G^\circ = -n_e F E^\circ_{\text{cell}}$ and $n_e = 6$,

$$-365 \times 1000 = -6 \times 96500 \times E^\circ_{\text{cell}}$$

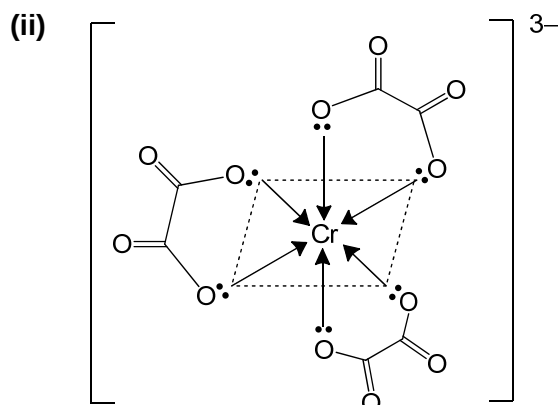
$$E^\circ_{\text{cell}} = +0.63 \text{ V} = E^\circ(\text{CrO}_4^{2-}/\text{Cr(OH)}_3) - (-0.76)$$

$$E^\circ(\text{CrO}_4^{2-}/\text{Cr(OH)}_3) = -0.13 \text{ V}$$



Since $E^\circ(\text{CrO}_4^{2-}/\text{X})$ is less positive than $E^\circ(\text{Cr}_2\text{O}_7^{2-}/\text{Cr}^{3+})$, it implies that CrO_4^{2-} is less readily reduced and thus CrO_4^{2-} is a weaker oxidising agent than $\text{Cr}_2\text{O}_7^{2-}$.

- (d) (i) reaction 1 : cold alkaline $\text{KMnO}_4(\text{aq})$
 reaction 2 : acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$, heat with immediate distillation
 reaction 3 : acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$, heat under reflux
 reaction 4 : acidified $\text{KMnO}_4(\text{aq})$, heat under reflux

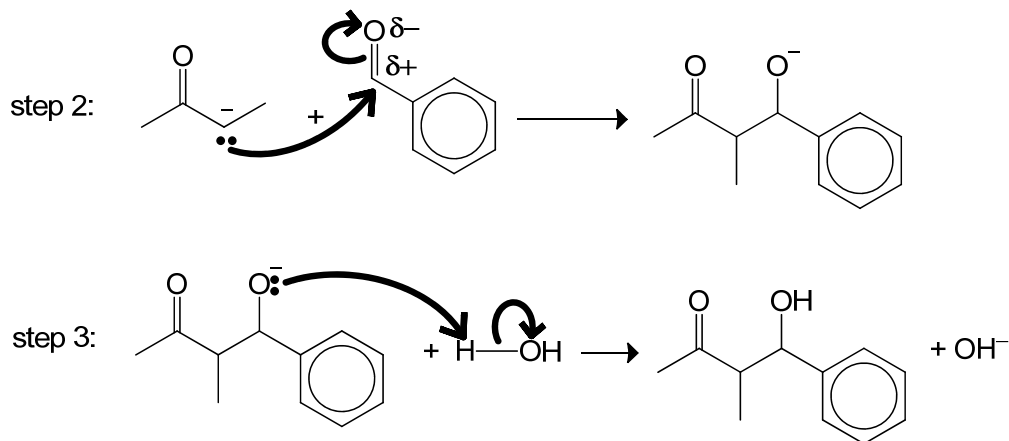


- (e) Heterogeneous catalysis.

When the reactant molecules adsorb on the catalyst surface, the bonds in the molecules are weakened, lowering the activation energy. The surface concentrations of the reactants are increased as well. The product formed then desorbs from the catalyst surface.

- 4 (a) (i) acid-base reaction
- (ii) p-p orbital overlap results in the delocalisation of lone pair on $\alpha\text{-C}^-$ into the adjacent $\text{C}=\text{O}$, dispersing the negative charge and thus making $\alpha\text{-anion}$ more stable than $\beta\text{-anion}$. Hence, $\alpha\text{-anion}$ is preferentially formed.
- (iii) nucleophile

(iv)



(b) (i)

bonds broken	
1 C=O	+740
2 C-H	2(+410)
Total	+1560

bonds formed	
1 C=C	+610
2 O-H	2(+460)
Total	+1530

$$\Delta H_r = +1560 - (+1530) = +30 \text{ kJ mol}^{-1}$$

(ii) Bond energies quoted from *Data Booklet* are average values derived from a full range of molecules that contain the particular bonds.

The bond energies quoted are for gaseous molecules but the reactants and products are in liquid state and hence, the enthalpy changes of vapourisation are not accounted for.

(iii)
$$\Delta G^\ominus = \underbrace{\Delta H^\ominus}_{+ve} - T \underbrace{\Delta S^\ominus}_{-ve}$$

When temperature increases, $-T\Delta S^\ominus$ becomes more negative.

So at high enough temperature, $\Delta G^\ominus < 0$ when $|\Delta H^\ominus| < |-T\Delta S^\ominus|$.

Hence, the Aldol Condensation is favoured by a high temperature.

(iv) Test: Add aqueous Br_2 to a small sample of the mixture.

Observation: If **P** is formed, orange $\text{Br}_2(\text{aq})$ will decolourise.

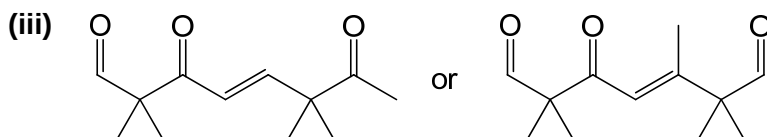
(c) (i) According to Lewis theory, an acid is an electron pair acceptor while a base is an electron pair donor.

(ii) Electron-donating $-\text{CH}_2\text{CH}_3$ makes the lone pair on O^- more available for protonation, making $\text{CH}_3\text{CH}_2\text{O}^-$ a stronger base than OH^- .

Hence, $\text{CH}_3\text{CH}_2\text{O}^-$ favour Aldol Condensation reaction.

(d) (i) 120°

(ii) The cyclisation of the above dicarbonyl compound would form a 4-membered ring and the bond angle of 90° is formed, resulting in angle strain in the ring and thus making the compound unstable.





Since C in CO is less electronegative than O in O₂, the lone pair on C in CO is more available for dative bond formation than O in O₂.

Hence, CO is a stronger ligand than O₂ and thus the formation of carboxyhaemoglobin occurs much more readily.

- (ii) During oxygen therapy, [O₂] is high and this causes the position of equilibrium 1 to shift right which reduces [Hb].

The reduction of [Hb] causes the position of equilibrium 2 to shift left which reduces [Hb(CO)₄].

Hence, the proportion of [Hb(O₂)₄] in the blood is increased while the proportion of Hb(CO)₄ in the blood is reduced, allowing more O₂ to be transported to other parts of the body.

- (b) (i) Any two of the following:

- Volume of gas molecules is negligible as compared to that of the gas.
- There are no intermolecular forces between the molecules.
- Collisions between molecules are perfectly elastic.
- Molecules move in constant, random motion.
- Kinetic energy of the molecules is directly proportional to the temperature.

- (ii) Since the molar ratio of CO : H₂O is 1:1, CO is the limiting reagent.

$$\text{Equilibrium amount of H}_2 = (2.0 \times 10^{-4}) \times \frac{97}{100} = 1.94 \times 10^{-4} \text{ mol}$$

Using ideal gas equation (*i.e.* $pV = nRT$),

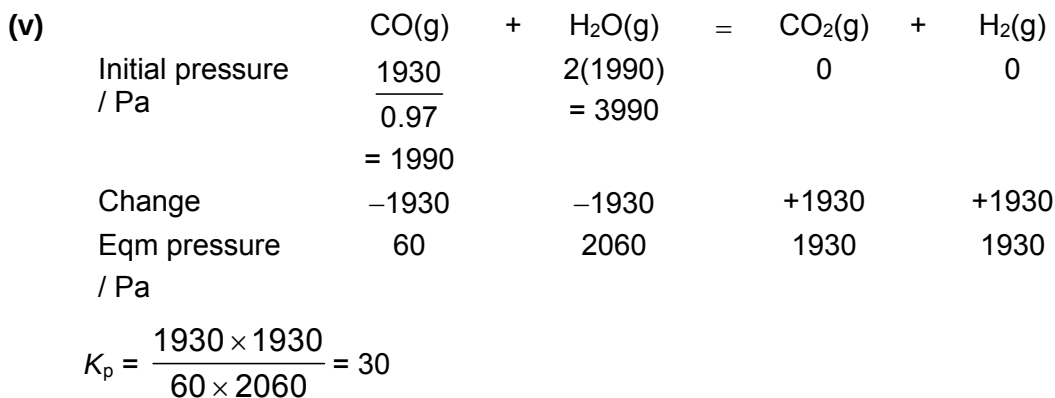
$$\begin{aligned} \text{equilibrium partial pressure of H}_2 &= \frac{(1.94 \times 10^{-4}) \times 8.31 \times 600}{500 \times 10^{-6}} \\ &= 1930 \text{ Pa} \end{aligned}$$

- (iii) It is valid as H₂ tends towards ideal gas behaviour since the H₂ molecule is small (or volume of the container is large due to low pressure) so the volume of the molecules is negligible as compared to that of the gas.

OR

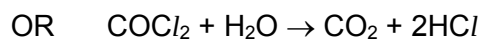
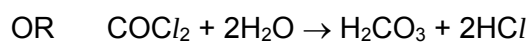
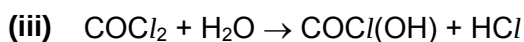
It is valid as H₂ tends towards ideal gas behaviour since the H₂ molecule has few electrons (or volume of the container is large due to low pressure and thus molecules are far apart or the molecules have high kinetic energies to overcome intermolecular forces due to high temperature) so the intermolecular forces in H₂ is negligible.

(iv) $K_p = \frac{p_{\text{CO}_2} \times p_{\text{H}_2}}{p_{\text{CO}} \times p_{\text{H}_2\text{O}}}$ Units of K_p : no units



- (c) (i) ΔS° is negative because there is a decrease in disorder of the system as the reaction proceeds with a decrease in amount of gaseous particles from 2 mol to 1 mol.

(ii) enthalpy driven



(d)

Observation	type of reaction	deduction
W is aromatic.	–	W contains benzene ring.
W (C ₁₂ H ₆ O ₁₂) + Na ₂ CO ₃ (aq) → CO ₂ (g)	acid-carbonate reaction	W is a carboxylic acid (or has –COOH).
1 mole of W gives 72 dm ³ CO ₂ gas	–	$2\text{RCOOH} + \text{Na}_2\text{CO}_3 \rightarrow 2\text{RCOONa} + \text{CO}_2 + \text{H}_2\text{O}$ amount of CO ₂ = $\frac{72}{24} = 3$ mol Since 2RCOOH ≡ 1CO ₂ , ∴ W has 6 –COOH.
W + appropriate reagent → sweet smelling X (C ₁₂ O ₉)	condensation	Recognise that 6 H and 3 O (<i>i.e.</i> 3H ₂ O) were lost when W is converted into X . X is a cyclic ester.

Combining the deductions, the structure of

